VI. Software Requirements Specification (SRS)

1. Introduction

1.1 Overview, Conventions and Intended Audience

The ABC Fashion AI Shopping Agent is a web-based prototype designed to support fashion product discovery through conversational interaction. The system enables customers to communicate their shopping needs in natural language and receive relevant product and outfit recommendations from the available catalogue.

This SRS defines the key functional, usability, performance, security, reliability and implementation requirements that guide the development and evaluation of the prototype.

This Software Requirements Specification (SRS) uses the IEEE SRS standard as a guide and follows the structure specified in the COMP3850 Project Deliverables Definition.

Requirements are identified using unique codes for consistency and traceability:

FR-XX – Functional Requirement

NFR-XX – Non-Functional Requirement

Shall/must indicate mandatory requirements, while should indicates desirable requirements. Requirements may be classified as Must Have, Nice to Have, or Out of Scope. Figures and tables will be numbered consistently.

This SRS is intended for the industry sponsor, project team, developers, testers, Quality Master, Scrum Master and other relevant stakeholders.

Developers can use the requirements to guide implementation, while testers and quality-assurance members can use them to develop test scenarios and acceptance criteria. The sponsor and project-management team can use the document to confirm that the solution remains aligned with the agreed MVP scope.

1.2 Purpose of Software Solution

The purpose of this Software Requirements Specification is to define the requirements for Version 1.0 of the ABC Fashion AI Shopping Agent.

The document establishes the expected system behaviour, constraints and quality requirements needed to develop, test and validate the MVP. It focuses on the conversational shopping and recommendation capabilities of the prototype and does not describe a complete production-ready e-commerce platform.

1.3 Scope of Project

The project will develop a conversational shopping agent that interprets customer requirements such as occasion, budget, style, size and colour, then retrieves and ranks suitable products from the ABC Fashion team-developed catalogue.

The MVP will support personalised product and coordinated outfit recommendations, conversational refinement, clarification of incomplete requests, cross-selling, upselling and fallback behaviour when an exact match cannot be identified. From a business perspective, the solution aims to improve product discovery and customer decision support while creating opportunities to present relevant complementary and higher-value products.

The prototype will use a team-developed fashion product dataset to support its recommendation logic.

The current MVP excludes real payment processing, order fulfilment, live retailer integration, virtual try-on, image or voice search, native mobile applications and production-grade deployment. Nice-to-Have capabilities may only be considered after the mandatory MVP requirements have been completed.

The MVP will also provide basic product catalogue administration functionality for authorised Product Administrators. Administrators will be able to add, update and remove product records, including product names, categories, prices, sizes, colours, availability and other catalogue information required by the shopping agent.

Figure 5 illustrates the system boundary and the main information exchanged between the customer, the ABC Fashion AI Shopping Agent and relevant external systems.

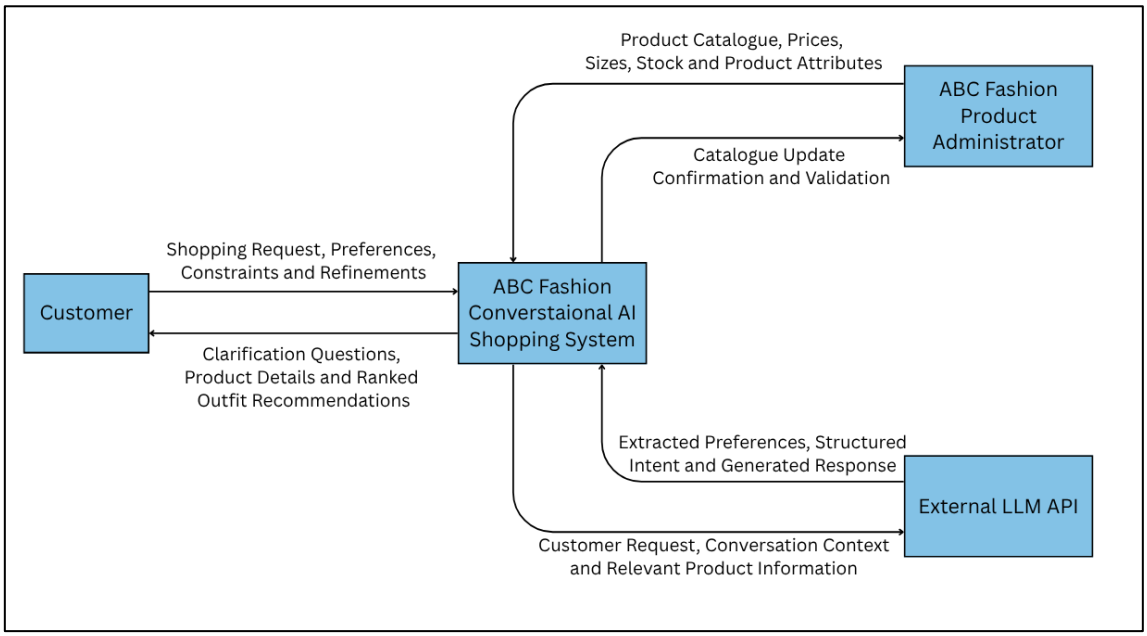


Figure 5: Proposed Customer-System Boundary for ABC Fashion AI Agent

The SRS may be refined throughout the Agile development process as requirements are clarified through development, testing and sponsor feedback. Approved changes will be recorded through the project's change-management process and revision history.

1.4 Definitions, Acronyms and Abbreviations

Group 20 will proceed with using the below acronyms and abbreviations as outlined in *Table 14*.

Table 14: Abbreviations and Definitions

Abbreviation	Definition
AI	Artificial Intelligence
MVP	Minimum Viable Product
FR-XX	Functional Requirement
NFR-XX	Non Functional Requirement
NFR-PXX	Non Functional Performance Requirement
NFR-SEXX	Non Functional Security Requirement

2. Overall Description

2.1 Product Perspective

The ABC Fashion Conversational AI Shopping Agent is a new, self-contained web-based prototype developed to improve product discovery within a simulated online fashion retail environment. It is not intended to replace an existing system or operate as a complete e-commerce platform; instead, it demonstrates how conversational AI can support customers in identifying suitable products and coordinated outfits.

The customer interacts with the system through a web interface, where shopping requirements are entered using natural language. The application processes these requirements, retrieves relevant information from the team-developed fashion product catalogue and uses an external Large Language Model (LLM) service to support intent interpretation and conversational responses. The resulting product recommendations are then presented back to the customer through the web interface.

The prototype is intentionally limited to conversational shopping and recommendation functionality.

2.2 Product Functions

The ABC Fashion Conversational AI Shopping Agent provides the following major functions:

- Accepts customer shopping requests through natural-language conversation.
- Identifies key preferences and constraints such as occasion, budget, style, size and colour.
- Searches and ranks products from the ABC Fashion product catalogue.
- Provides personalised product and coordinated outfit recommendations.
- Supports cross-selling through complementary product suggestions.
- Supports upselling through relevant higher-value alternatives.
- Maintains conversation context and allows customers to refine previous requests.
- Asks clarification questions when customer requirements are incomplete or unclear.
- Provides appropriate fallback responses when suitable products cannot be identified.
- Presents recommendations through a web-based shopping interface.
- Allows authorised Product Administrators to add products to the product catalogue.
- Allows authorised Product Administrators to update existing product information.
- Allows authorised Product Administrators to remove products from the catalogue.
- Ensures catalogue changes are reflected in subsequent product searches and recommendations.

Figure 6 shows how the conversational shopping system connects the customer, web interface, product database and external LLM service to generate relevant product recommendations and responses.

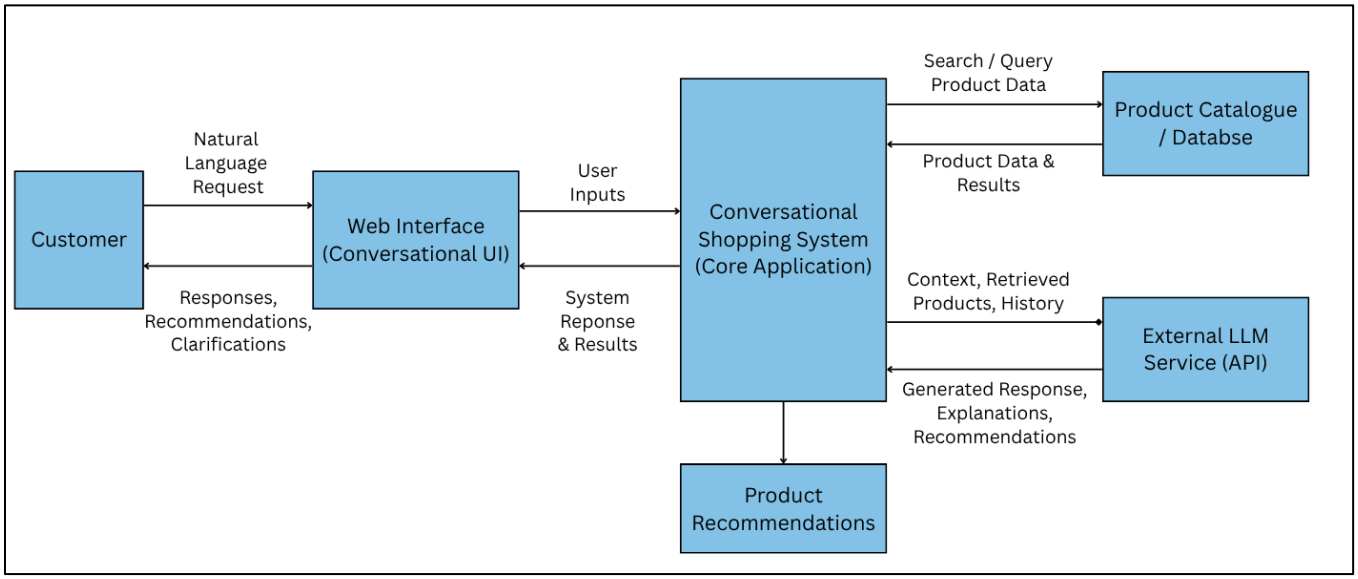


Figure 6: Connections and Workflows for ABC Fashion's proposed AI Agent

2.3 User Classes and Characteristics

2.3.1 Customer

Customers are the primary users of the ABC Fashion AI Shopping Agent and are general online fashion shoppers with no technical knowledge required. They can describe preferences such as product type, occasion, budget, style, colour and size through natural-language conversation and refine these preferences as needed.

2.3.2 Project Testers and Evaluators

Project team members and testers will use representative shopping scenarios to assess system functionality, recommendation quality, usability and reliability against the SRS requirements.

2.4 Operating Environment

The system will operate as a web-based application on standard internet-connected desktop, tablet and mobile devices (see *Table 15*).

Table 15: System Architecture and Chosen Applications

System	Application
Frontend	Next.js / React
Backend/Server Side App	Node.js / Next.js
Database	Supabase / PostgreSQL
AI Service	External LLM API, such as OpenAI or Gemini
Hosting	Vercel or equivalent cloud platform
Supported Browsers	Chrome, Safari and Edge

Version Control	Github

AI processing and product retrieval will primarily occur through server-side and cloud-based services, so no specialised hardware or software installation is required.

2.5 User Documentation

As the AI shopping agent is designed to be intuitive, only brief user guidance will be provided. This will include a short introduction, example prompts, and clear error, clarification and no-match messages. Authorised Product Administrators will also receive basic guidance for maintaining the product catalogue. All documentation will be delivered electronically as part of the prototype.

3. Requirements

This section defines the functional, design, implementation, usability and non-functional requirements of the ABC Fashion Conversational AI Shopping Agent. The requirements describe the capabilities and constraints necessary for the prototype to help customers identify suitable fashion products through natural-language conversation, personalised recommendations, conversational refinement, cross-selling and upselling. These capabilities directly reflect the sponsor's objective of exploring how a conversational AI agent can help online retail customers find suitable products more quickly.

3.1 Functional Requirements

The functional requirements define the behaviour that the system shall provide to customers. Requirements are grouped according to the major functions of the conversational shopping experience.

3.1.1 Conversational Shopping and Preference Understanding

Conversational recommender systems support interactive preference elicitation, allowing customers to express their needs through dialogue and enabling the system to progressively obtain additional preference information through clarification and feedback [1], [2], [3]. The customer based functional requirements associated with the proposed AI Agent are stipulated in *Table 16* below.

Table 16: Customer Based Functional Requirements

ID	Functional Requirement
FR-01	The system shall allow customers to submit natural-language shopping requests.
FR-02	The system shall extract any customer-provided occasion, style, colour, size, product type and budget preferences from the shopping request.
FR-03	The system shall distinguish between preferences explicitly provided by the customer and information that has not been provided.
FR-04	The system shall ask a clarifying question when a required preference or constraint cannot be determined from the customer's request and is necessary to retrieve suitable products.
FR-05	The system shall avoid requesting information that the customer has already provided during the current conversation.

FR-06	When the system cannot reliably interpret a shopping request, it shall inform the customer that clarification is required and prompt the customer to rephrase or provide additional information.

A typical interaction begins when the customer enters a request such as *"I need an outfit for a beach wedding under $250."* The system analyses the request, extracts the relevant preferences and asks follow-up questions only where necessary.

3.1.2 Product Search and Personalised Recommendations

The system shall use identified customer requirements to retrieve relevant products from the ABC Fashion product catalogue and provide personalised recommendations.

The recommendation based functional requirements associated with the proposed AI Agent are stipulated in *Table 17* below.

Table 17: Recommendation Based Functional Requirements

ID	Functional Requirement
FR-07	The system shall retrieve products from the available ABC Fashion product catalogue.
FR-08	The system shall filter catalogue products using each explicit customer constraint that corresponds to an available catalogue attribute.
FR-09	The system shall prioritise products that satisfy all identified hard constraints, then rank remaining products according to the number of identified soft preferences they satisfy.
FR-10	The system shall consider explicit customer budget constraints when generating recommendations.
FR-11	The system shall display the product image, product name, price and available sizes for each recommended product where those attributes exist in the catalogue.
FR-12	The system shall provide a concise explanation of why each recommended product matches the customer's request.
FR-13	The system shall not invent products or product attributes that are unsupported by the available catalogue data.
FR-14	When no product satisfies all identified preferences, the system shall search for alternatives by relaxing one or more soft preferences, such as colour or style, while continuing to satisfy hard constraints such as required size and maximum budget. The system shall inform the customer which stated preferences are not satisfied by each alternative.

The Salesforce project brief specifically identifies understanding styling intent, recommending relevant products, and identifying cross-sell and upsell opportunities as core objectives of the proposed agent. Providing explanations for recommendations can improve their transparency and user acceptance [4]. However, LLM-based conversational recommenders may generate unsupported or misleading information, making it important to ground product recommendations and explanations in available catalogue data [5], [6], [7].

3.1.3 Conversational Refinements and Styling Recommendations

The system shall maintain relevant conversational context so customers can progressively refine their shopping requirements without restarting the interaction.

A defining characteristic of conversational recommender systems is their ability to support multi-turn interactions in which users progressively refine preferences and provide feedback on previous recommendations [1], [2].

The conversation based functional requirements associated with the proposed AI Agent are stipulated in *Table 18* below.

For example, after receiving an initial recommendation, a customer may respond with *"Something cheaper"* or *"Can you make it blue instead?"*. The system shall update the affected preference while retaining the other relevant constraints.

Table 18: Conversation Based Functional Requirements

ID	Functional Requirement
FR-15	The system shall retain the customer's identified product type, occasion, style, colour, size and budget preferences throughout the active shopping conversation until they are modified or the conversation is reset.
FR-16	The system shall allow customers to modify previously stated preferences using follow-up messages.
FR-17	The system shall retain previously provided preferences unless the customer modifies or contradicts them.
FR-18	The system shall prioritise the customer's most recent explicit preference when conflicting information is provided.
FR-19	The system shall allow customers to request alternatives based on a stated change, including a lower price, different colour, different style or similar product type.
FR-20	When a customer accepts, rejects or requests a change to a previous recommendation, the system shall apply that feedback when generating the next set of recommendations.
FR-21	The system shall generate coordinated outfit recommendations using products from two or more compatible product categories while satisfying the customer's explicit size and maximum-budget constraints.
FR-22	The system shall provide cross-sell recommendations by suggesting products from a complementary product category that satisfy the customer's explicit constraints.

FR-23	The system shall present an optional upsell when a higher-priced product from the same or equivalent product category satisfies the customer's explicit requirements and provides an alternative to the original recommendation.
FR-24	The system shall not recommend an upsell that violates an explicit maximum customer budget unless the customer agrees to modify that constraint.

3.1.4 Web Shopping Experience and Error Handling

The conversational agent shall be accessible through the ABC Fashion web-based shopping prototype.

The experience based functional requirements associated with the proposed AI Agent are stipulated in *Table 19* below.

Table 19: Experience Based Functional Requirements

ID	Functional Requirement
FR-25	The system shall provide access to the conversational shopping assistant through the ABC Fashion web application.
FR-26	The system shall provide an interface for entering and displaying conversational messages.
FR-27	The system shall visually present recommended products within the shopping interface.
FR-28	The system shall allow customers to continue interacting with the agent after recommendations have been displayed.
FR-29	When a technical error prevents a request from being completed, the system shall display a user-facing error message stating that the request could not be completed and provide an option to retry or modify the request.
FR-30	The system shall allow customers to modify their request when no suitable products are found.
FR-31	The system shall allow customers to open the product page of a recommended product.
FR-32	The system shall allow customers to add an available recommended product to their shopping cart.
FR-33	The system shall display the current availability of recommended product sizes where catalogue availability data is provided.

3.1.5 Product Catalogue Administration

The system shall provide authorised Product Administrators with basic functionality for maintaining the product catalogue used by the conversational shopping agent.

The administration based functional requirements associated with the proposed AI Agent are stipulated in *Table 20* below.

Table 20: Administration Based Functional Requirements

ID	Functional Requirement
FR-34	The system shall allow an authorised Product Administrator to add a new product to the product catalogue.
FR-35	The system shall allow an authorised Product Administrator to update an existing product record.
FR-36	The system shall allow an authorised Product Administrator to remove an existing product from the product catalogue.
FR-37	The system shall allow administrators to maintain product information including product name, category, price, available sizes, colours, availability and product image or URL where applicable.
FR-38	The system shall validate required product fields before creating or updating a product record.
FR-39	The system shall restrict product catalogue administration functions to authorised Product Administrators.
FR-40	Changes made to the product catalogue shall be reflected in subsequent product searches and recommendations.

3.2 Design and Implementation Requirements/Constraints

The following requirements specify constraints on the technologies, architecture, interfaces and implementation of the prototype. These constraints support the development of a maintainable web-based solution while protecting credentials and ensuring compatibility between the application's components.

3.2.1 Web Application Architecture

The system shall be implemented as a web-based application consisting of a customer-facing interface, server-side application logic, product database and external AI service.

The proposed implementation will use:

- **Next.js and React** for the customer-facing web application;
- **Next.js/Node.js server-side functionality** for application and recommendation logic;
- **Supabase/PostgreSQL** for structured product catalogue storage;
- **OpenAI or Gemini API** for natural-language interpretation and conversational response generation;
- **CSV, JSON or equivalent structured formats** for initial product-data import;
- **GitHub** for source-code version control; and
- **Vercel or an equivalent cloud-hosting service** for deployment.

The final AI provider and supporting technologies may be adjusted during development where testing or sponsor feedback demonstrates that an alternative better satisfies the project's requirements.

3.2.2 Software Interfaces

The system shall support communication between the following components listed in *Table 21*:

Table 21: Component Communications

Software Component	Purpose
Next.js / React Web Application	Provides the customer-facing shopping and conversational interface.
Next.js / Node.js Server	Processes requests, coordinates product retrieval and communicates with the AI service.
Supabase / PostgreSQL	Stores and retrieves structured product catalogue information.
OpenAI or Gemini API	Interprets natural-language shopping requests and assists with conversational responses and recommendations.
Product Dataset	Provides product catalogue records imported into the database.
Vercel or Equivalent Hosting	Hosts the deployed web prototype.

The primary information exchanged between these components shall include customer requests, extracted preferences, structured product-search criteria, product records, AI context and AI-generated conversational responses.

Structured JSON shall be used for application-to-service communication where appropriate

3.2.3 Communications Constraints

The customer's browser, application server, database and AI provider will communicate through internet-based interfaces.

All external communications containing application data shall use secure encrypted connections, preferably HTTPS using TLS 1.2 or later.

AI API requests and database operations shall be performed through server-side application logic rather than directly from the customer's browser.

API credentials, database credentials and other application secrets shall be maintained using secured server-side environment variables.

The system shall not require FTP, peer-to-peer communication or direct device-to-device communication for the prototype.

3.2.4 Hardware and Platform Constraints

No specialised hardware shall be required.

The application shall operate using standard internet-connected:

- desktop computers;
- laptop computers;
- smartphones; and
- tablets.

Users shall interact with the system primarily using a keyboard, mouse, trackpad or touchscreen.

AI processing, recommendation generation and product retrieval shall primarily occur server-side. Customer devices therefore only require a modern web browser and stable internet connection.

The prototype will not directly interface with barcode scanners, payment terminals, cameras, microphones, biometric devices or Internet of Things devices.

3.2.5 Data and AI Constraints

Product recommendations shall be grounded in the available ABC Fashion product dataset. This is important because LLM-based recommender systems can produce unsupported product information or recommend items outside the available item domain [5], [6], [7].

The AI model shall not be treated as the authoritative source for factual product information such as:

- product name;
- price;
- size;
- colour;
- availability;
- product category; or
- product URL.

These attributes shall originate from the available catalogue data.

Only information necessary to interpret the customer's request and generate recommendations should be transmitted to the external AI service.

Payment information and unnecessary sensitive customer information shall not be transmitted to the AI provider.

3.2.6 Development and Version Control

Exact software dependency versions shall be recorded and controlled through the project's package configuration and Git repository.

The frontend, product catalogue, recommendation logic and AI integration should be implemented as logically separated components to support modification and testing.

The interfaces between the database and AI provider shall use clearly defined APIs and data structures so that these components can be replaced or extended without requiring a complete redesign of the application.

3.3 Usability Requirements

The system shall provide a simple and intuitive conversational shopping experience for customers who may have no prior knowledge of AI systems. Research on preference elicitation in conversational

recommender systems indicates that the way systems request, clarify and refine user preferences can affect the overall user experience [3]. Usability requirements are listed in *Table 22* below:

Table 22: Usability Requirements

ID	Usability Requirement
UR-01	A new customer shall be able to begin a shopping conversation without requiring training or documentation.
UR-02	Customers shall be able to express common shopping requirements using ordinary natural language rather than predefined commands.
UR-03	The interface shall visually distinguish customer messages, AI responses and product recommendations using separate interface elements or labels.
UR-04	Each recommended-product card shall display the product name, image, price and available sizes where these attributes are present in the catalogue.
UR-05	Customers shall be able to refine preferences through follow-up conversation without restarting the shopping session.
UR-06	The system shall provide visible feedback, such as a loading indicator, while a recommendation or response is being generated.
UR-07	Error messages shall use non-technical language and, where recovery is possible, provide a retry or request-modification action.
UR-08	When information required to perform a product search is missing or cannot be interpreted, the system shall ask the customer for the missing or unclear information rather than terminate the interaction.
UR-09	The interface shall provide a clear mechanism for starting a new conversation or resetting the current shopping context.
UR-10	The interface shall allow all core shopping-agent functions to be used on desktop, tablet and mobile screen sizes without horizontal page scrolling or loss of required controls.
UR-11	Interactive interface elements shall use descriptive labels and support keyboard interaction where applicable.
UR-12	Each product image displayed to the customer shall include alternative text containing at least the product name or product type.

UR-13	The interface shall use consistent typography, spacing and control styles across the shopping and conversational interfaces, and all text and controls shall remain visible against their backgrounds.
UR-14	The system shall display between 3 and 6 products in the initial recommendation set where at least 3 matching catalogue products are available.

The interface will follow a clean, modern fashion e-commerce design. The primary interface will consist of a conversational chat area and visual product recommendations, enabling the customer to move naturally between describing their requirements, viewing products and refining the results.

Example user-facing error messages should avoid exposing technical implementation details. For example: "We couldn't generate recommendations right now. Please try again."

3.4 Other Non-Functional Requirements

3.4.1 Performance Requirements

The system shall provide sufficiently responsive interactions so that the conversational agent improves rather than increases friction within the shopping process. As such, the performance requirements for the AI Agent have been provided in *Table 23* below:

Table 23: Performance Requirements

ID	Performance Requirement
NFR-P01	During performance testing in the team's documented test environment, the main shopping interface shall load within 3 seconds for at least 95% of test runs.
NFR-P02	During performance testing in the team's documented test environment, product catalogue search and filtering operations shall return results within 2 seconds for at least 95% of requests.
NFR-P03	The system shall display either an AI response or a visible processing indicator within 5 seconds of the customer submitting a request.
NFR-P04	The system shall support at least 20 concurrent prototype users without significant degradation of core functionality.
NFR-P05	While an external AI or database request is pending, the system shall display a processing indicator. If the request fails or reaches the configured service timeout, the system shall display an error message and allow the customer to retry.

Performance shall be evaluated using representative scenarios including occasion-based searches, budget-constrained recommendations, outfit generation and cross-selling.

3.4.2 Security and Privacy Requirements

Security is particularly important because the application communicates with an external AI service and database while processing customer-provided conversational information. Recommender systems may process user preference and behavioural information, creating security and privacy risks associated with unauthorised access, disclosure and misuse of user data [9]. To ensure all security requirements are met, explicit guidelines have been outlined in *Table 24* below:

Table 24: Security Requirements

ID	Security Requirement
NFR-SE01	All application-managed communication between the customer's browser, application server, database and external AI service shall use HTTPS with TLS 1.2 or later.
NFR-SE02	API keys, database credentials and other secrets shall not be stored in client-side code or committed to the public source-code repository.
NFR-SE03	Application credentials shall be managed using secured server-side environment variables.
NFR-SE04	Database access shall follow the principle of least privilege.
NFR-SE05	User input shall be validated and sanitised before being processed or incorporated into database operations.
NFR-SE06	Anonymous shopping and recommendation functionality shall not require the customer to provide a name, email address, postal address, payment information or other account-identifying information.
NFR-SE07	Conversation content shall not be permanently stored after the shopping session ends unless storage is enabled for a documented and approved testing, evaluation or system feature.
NFR-SE08	If customer history or personalised account functionality is implemented, users shall be authenticated before accessing account-specific information.
NFR-SE09	Anonymous product search and recommendation functionality shall not require authentication.
NFR-SE10	Application logs shall not contain API keys, passwords or -unnecessary personal information.
NFR-SE11	Before prototype deployment, the application shall be checked for unsafe input handling, exposed API/database credentials and unauthorised database or catalogue access.
NFR-SE12	Information transmitted to an external AI service shall be limited to information necessary to process the shopping request.
NFR-SE13	Payment-card details and other unnecessary sensitive information shall not be transmitted to the AI service.

Where personal information is collected, the prototype should follow applicable Australian privacy requirements and data-minimisation principles.

The inclusion of these controls also reflects the sponsor material's emphasis on a trust layer surrounding the models, applications and contextual data used within an agentic system.

3.4.3 Safety and Responsible AI Requirements

Although the prototype does not control physical equipment, risks can arise from inappropriate, misleading or unsupported AI-generated responses. LLM-based conversational recommender systems can produce unsupported explanations or recommendations, making catalogue grounding and explicit handling of uncertainty important safeguards [5], [6], [7].

Responsible AI requires ethical principles such as fairness, transparency and accountability to be translated into concrete system practices and governance mechanisms rather than treated only as high-level principles [10] (see *Table 25)*.

Table 25: Ethical Requirements

ID	Ethical Requirement
NFR-S01	The AI agent shall reject or redirect requests outside the intended fashion-shopping domain when they involve harmful, offensive or discriminatory content, and shall pass the team's defined safety test scenarios without producing prohibited responses.
NFR-S02	Product information presented to customers shall be based on available catalogue data rather than invented by the AI model.
NFR-S03	When a requested product attribute is missing from the catalogue data or no catalogue record supports an AI-generated product claim, the system shall state that the information is unavailable rather than presenting the claim as fact.
NFR-S04	The prototype shall not independently perform purchases, payments or other irreversible customer transactions without explicit customer action.
NFR-S05	Customers shall remain able to reject, modify or refine recommendations throughout the interaction.

These safeguards are particularly relevant because the sponsor defines an AI agent as software that uses an LLM to reason, plan and use tools towards completing a customer goal.

3.4.4 Reliability and Robustness Requirements

The reliability and robustness requirements for the AI Agent have been outlined in *Table 26*.

Table 26: Reliability Requirements

ID	Reliability Requirement
NFR-R01	Failure of an AI request, product search or database operation shall not cause the entire application to crash.
NFR-R02	When a recoverable AI-service, database or product-search failure occurs, the system shall display a non-technical error message and provide an option to retry the request.
NFR-R03	Customers shall be able to retry an unsuccessful request.
NFR-R04	When a shopping request is incomplete or ambiguous, the system shall request the missing or unclear information; when the request cannot be processed, the system shall instruct the customer to rephrase it.
NFR-R05	Product details shown to the customer shall remain consistent with the underlying catalogue data.
NFR-R06	Recommendations shall satisfy all explicit hard customer constraints for which corresponding catalogue attributes are available, including maximum budget and required size.

3.4.5 Maintainability and Adaptability Requirements

The maintainability and adaptability requirements for the AI Agent have been outlined in *Table 27*.

Table 27: Maintainability and Adaptability Requirements

ID	Adaptability Requirement
NFR-M01	Frontend, catalogue/database, recommendation and AI-integration functionality shall be separated into logical components.
NFR-M02	Product catalogue records shall be addable, editable and removable without requiring changes to the application's source code or architecture.
NFR-M03	Recommendation prompts and AI configuration should be modifiable without redesigning unrelated application components.
NFR-M04	The external AI provider shall be replaceable by modifying the AI-integration component and associated configuration without requiring changes to the customer-facing interface or product-database structure.
NFR-M05	Source code and configuration changes shall be managed through version control.

3.4.6 Compatibility and Portability Requirements

The compatibility and portability requirements for the AI Agent have been outlined in *Table 28*.

Table 28: Compatibility and Portability Requirements

ID	Compatibility Requirement
NFR-C01	The web application shall support the latest stable desktop versions of Chrome, Safari and Edge available at the beginning of system testing.
NFR-C02	The web application shall support the latest stable versions of Safari on iOS and Chrome on Android available at the beginning of system testing.
NFR-C03	The application shall not require customers to install additional software.
NFR-C04	All functions defined in FR-01–FR-40 shall remain accessible on supported desktop, tablet and mobile devices, except administrator functions that are intentionally restricted to supported administrator interfaces.

3.4.7 Availability Requirements

NFR-A01: During scheduled prototype demonstrations and testing, the deployed application should remain accessible for at least 99% of the planned testing period, excluding planned maintenance and outages caused by external third-party services.

For the prototype, usability, correctness, security and reliability will take priority over unnecessary complexity. A smaller set of relevant recommendations that accurately reflects the customer's requirements and the available catalogue is preferable to a more sophisticated agent that generates inconsistent or unsupported results. This is consistent with the project's focus on building a proof of concept and evaluating it through realistic customer scenarios and recommendation quality.

IX. Analysis Documentation

1. Use Case Diagram

The below diagram depicts the current use cases associated with the ABC Fashion AI Shopping Assistant platform (Figure 7).

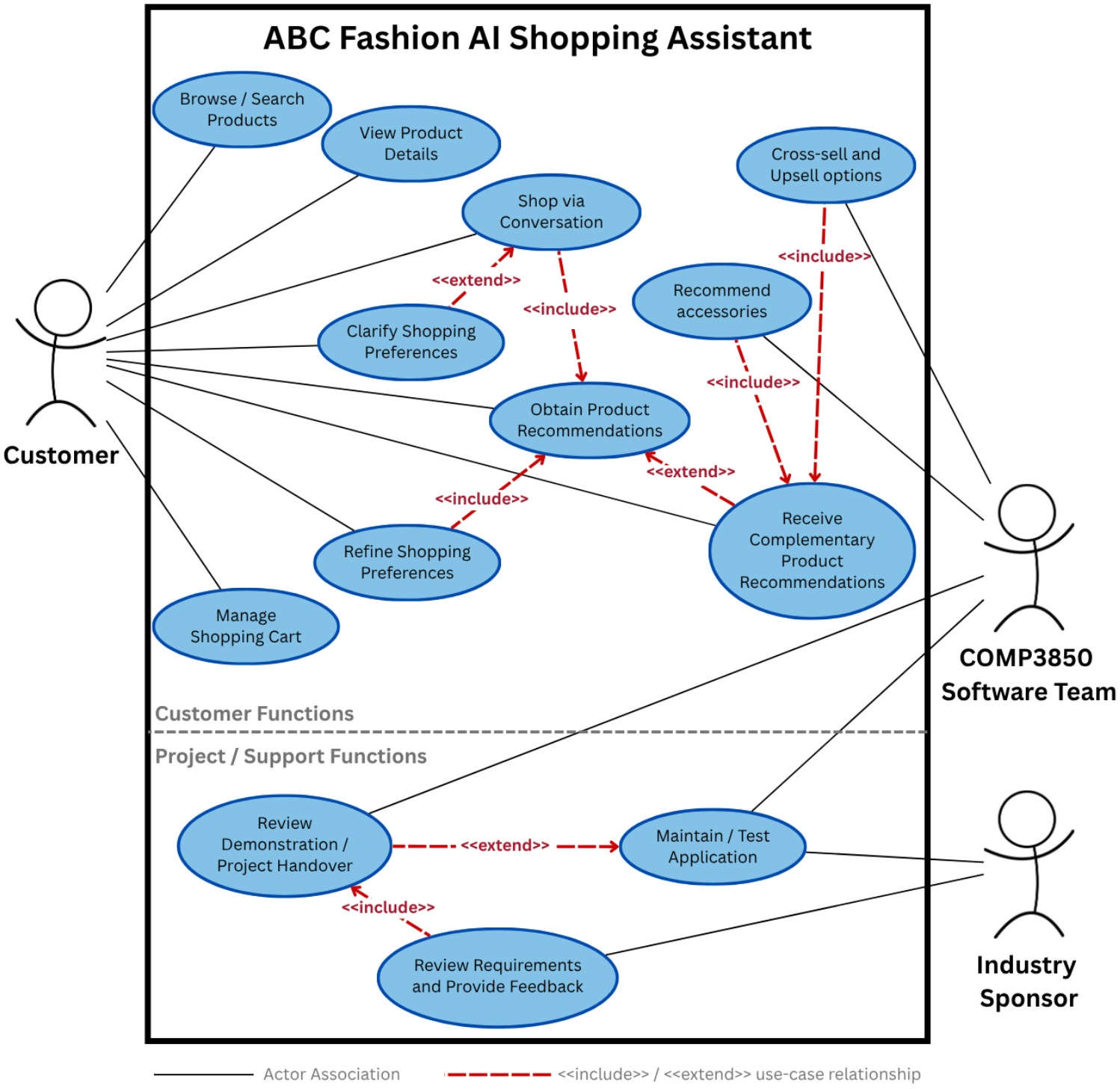


Figure 7: ABC Fashion Use Case Diagram

2. Use Case Descriptions

The below tables contain various use case descriptions relevant to the ABC Fashion AI Shopping Assistant.

UC01 - Shop via Conversation

Table 32: UC01 – Shop via Conversation

Use Case	UC01 - Shop via Conversation
Goal	Allow a customer to communicate their shopping requirements using natural language and receive assistance from the ABC Fashion AI Shopping Agent in identifying suitable products.
Preconditions	The ABC Fashion web application and conversational shopping interface are available to the customer.
Success End Condition	The customer's shopping requirements have been interpreted and suitable product recommendations have been presented.
Failed End Condition	The customer's request cannot be processed and no recommendations are produced. The customer is informed of the problem and may retry or modify the request.
Primary Actors	Customer
Secondary Actors	None
Trigger	The customer submits a natural-language shopping request to the AI shopping agent.

Description / Main Success Scenario

Table 33: UC01 Description and Success Scenarios

Step	User Action	Step	System Response
1	Customer opens the conversational shopping interface.	2	System displays the shopping conversation interface.
3	Customer enters a shopping request describing one or more requirements, such as product type, occasion, budget, size, colour or style.	4	System receives and analyses the request.
5	-	6	System identifies the preferences and constraints explicitly provided by the customer.
7	-	8	System determines that sufficient information exists to search for suitable products.

9	-	10	System performs UC03 - Obtain Product Recommendations.
11	Customer reviews the recommendations and may continue the conversation.	12	System maintains the active conversation and relevant shopping preferences.

Alternative Flows

Table 34: UC01 – Alternative Flows

Step	Branching Action - User	Branching Action - System
A1	Customer provides insufficient information.	System performs UC02 - Clarify Shopping Preferences.
A2	Customer submits a request that cannot be reliably interpreted.	System informs the customer and asks them to rephrase or provide additional information.
A3	Customer attempts to submit a request while a technical failure occurs.	System displays an error and provides an option to retry or modify the request.

UC02 - Clarify Shopping Preferences

Table 34: UC02 – Clarify Shopping Preferences

Use Case	UC02 - Clarify Shopping Preferences
Goal	Obtain additional information from the customer when a preference or constraint required to identify suitable products is missing or unclear.
Preconditions	An active shopping conversation exists and the customer has submitted a shopping request that does not contain sufficient interpretable information.
Success End Condition	The missing or unclear preference has been obtained and added to the customer's current shopping context.
Failed End Condition	The required preference remains unclear or cannot be obtained, preventing the system from reliably performing the intended product search.
Primary Actors	Customer
Secondary Actors	None
Trigger	During UC01, the system determines that a required preference or constraint cannot be determined from the customer's request.

Description / Main Success Scenario

Table 36: UC02 Descriptions and Success Scenarios

Step	User Action	Step	System Response
1	-	2	System identifies a required preference or constraint that is missing or unclear.
3	-	4	System asks the customer an appropriate clarification question.
5	Customer provides the requested information.	6	System interprets the customer's response.
7	-	8	System updates the current shopping preferences with the newly provided information.
9	-	10	System returns to the shopping conversation and continues product discovery.

Alternative Flows

Table 37: UC02 – Alternative Flows

Step	Branching Action - User	Branching Action - System
A1	The customer provides another unclear response.	The system asks the customer to rephrase or provide additional information.
A2	The customer provides information that conflicts with a previous preference.	The system treats the customer's latest explicit preference as the current preference.
A3	The customer supplies information that was already known.	The system retains the information without requesting it again.
A4	The customer does not know or does not wish to provide a preference.	The system proceeds with UC03 using the available context and allows the customer to refine the results later through UC06.

UC03 - Obtain Product Recommendations

Table 38: UC03 – Obtain Product Recommendations

Use Case	UC03 - Obtain Product Recommendations
Goal	Identify and present catalogue products that best satisfy the customer's shopping requirements and preferences.
Preconditions	The system has enough context to identify at least one relevant product category, and the ABC Fashion product catalogue is available to the system.
Success End Condition	One or more suitable catalogue products are presented to the customer with relevant product information and recommendation explanations.
Failed End Condition	No suitable products or acceptable alternatives can be identified, and the customer is informed that the request should be modified.
Primary Actors	Customer
Secondary Actors	None
Trigger	Identify and present catalogue products that best satisfy the customer's shopping requirements and preferences.

Description / Main Success Scenario

Table 39: UC03 – Descriptions and Success Scenarios

Step	User Action	Step	System Response
1	Customer provides sufficient shopping preferences through the conversation.	2	System converts the identified preferences into product-search criteria.
3	-	4	System retrieves candidate products from the ABC Fashion catalogue.
5	-	6	System filters products according to explicit customer constraints such as budget and size.
7	-	8	System ranks the remaining products according to how well they satisfy the customer's preferences.
9	-	10	System selects suitable recommendations.
11	-	12	System displays available product information such as image, name, price and available sizes.

13	Customer reviews the recommendations.	14	System provides a concise explanation of why each recommended product matches the request.

Alternative Flows

Table 40: UC03 – Alternative Flows

Step	Branching Action - User	Branching Action - System
A1	The customer's complete set of preferences cannot be satisfied.	The system relaxes one or more soft preferences while continuing to respect hard constraints such as required size and maximum budget.
A2	The customer receives an alternative that does not satisfy every soft preference.	The system identifies which stated preferences are not satisfied by the alternative.
A3	The customer's request produces no suitable alternatives.	The system informs the customer and allows the request to be modified.
A4	The customer has not provided a size or budget.	The system presents recommendations without applying those filters. Size selection is deferred until the customer adds a product to the cart (UC07).

UC04 - Receive Complementary Product Recommendations

Table 41: UC04 – Receive Complimentary Product Recommendations

Use Case	UC04 - Obtain Complementary Product Recommendations
Goal	Provide the customer with relevant coordinated, cross-sell or upsell product recommendations that complement the products already being considered.
Preconditions	The customer has received at least one suitable product recommendation and relevant complementary catalogue products exist.
Success End Condition	Appropriate complementary or alternative products are presented while respecting the customer's applicable constraints.
Failed End Condition	No suitable complementary products are available, so no additional recommendation is presented.
Primary Actors	Customer
Secondary Actors	None
Trigger	A suitable primary recommendation has been generated and an appropriate complementary recommendation opportunity exists.

Description / Main Success Scenario

Table 42: UC04 – Descriptions and Success Scenarios

Step	User Action	Step	System Response
1	Customer receives or reviews a primary product recommendation.	2	System evaluates the recommended product and the customer's current preferences.
3	-	4	System searches for compatible products from relevant complementary categories.
5	-	6	System checks candidate products against explicit customer constraints.
7	-	8	System selects suitable complementary recommendations.
9	-	10	System presents the complementary products to the customer.
11	Customer reviews the additional recommendations.	12	System maintains the current conversation and shopping context.

Alternative Flows

Table 43: UC04 – Alternative Flows

Step	Branching Action - User	Branching Action - System
A1	The customer is seeking a coordinated outfit.	The system recommends products from two or more compatible product categories while respecting size and budget constraints.
A2	The customer is viewing a product suitable for cross-selling.	The system suggests an appropriate product from a complementary category.
A3	The customer asks for a better or higher-quality option.	The system may present a higher-priced alternative only when it offers a clear benefit and still satisfies the customer's requirements.
A4	The customer has specified a maximum budget.	The system does not present an upsell exceeding that budget unless the customer agrees to modify the constraint.
A5	-	If no suitable complementary product exists, the system continues without presenting one.

UC05 - View Product Details

Table 44: UC05 – View Product Details

Use Case	UC05 - View Product Details
Goal	Allow the customer to view additional catalogue information about a recommended product.
Preconditions	At least one product has been displayed to the customer and a product page is available for the selected item.
Success End Condition	The selected product's details are displayed to the customer.
Failed End Condition	The selected product page cannot be opened or its information cannot be retrieved.
Primary Actors	Customer
Secondary Actors	None
Trigger	Customer selects a displayed recommended product.

Description / Main Success Scenario

Table 45: UC05 – Descriptions and Success Scenarios

Step	User Action	Step	System Response
1	Customer reviews displayed product recommendations.	2	System displays selectable recommended products.
3	Customer selects a product they wish to inspect.	4	System retrieves the selected product's available catalogue information.
5	-	6	System opens the selected product page.
7	-	8	System displays available details such as product image, name, price, size and availability.
9	Customer reviews the product information.	10	System continues to make the shopping interface available for further actions.

Alternative Flows

Table 46: UC05 – Alternative Flows

Step	Branching Action - User	Branching Action - System
A1	The customer selects a product whose availability has changed.	The system displays the current availability provided by the catalogue.

A2	The customer selects a product with incomplete catalogue data.	The system displays only available catalogue-supported information rather than inventing missing details.
A3	The customer selects a product that cannot currently be retrieved.	The system informs the customer that the product information cannot be displayed.
A4	The customer selects a product with no matching product image.	The system displays a branded placeholder image together with the available catalogue information.

UC06 - Refine Shopping Preferences

Table 47: UC06 – Refine Shopping Preferences

Use Case	UC06 - Refine Shopping Preferences
Goal	Allow the customer to modify previously stated shopping preferences through follow-up conversation without starting a new shopping session.
Preconditions	An active shopping conversation exists and the system currently holds one or more customer preferences.
Success End Condition	The relevant preference is updated, unchanged preferences are retained, and new recommendations are generated using the revised requirements.
Failed End Condition	The customer's requested refinement cannot be interpreted and the system cannot reliably update the shopping preferences.
Primary Actors	Customer
Secondary Actors	None
Trigger	Customer submits a follow-up message that changes a preference or gives feedback about previous recommendations.

Description / Main Success Scenario

Table 48: UC06 – Descriptions and Success Stories

Step	User Action	Step	System Response
1	Customer reviews the current recommendations.	2	System maintains the customer's current shopping preferences and conversational context.
3	Customer submits a refinement such as "something cheaper" or "make it blue."	4	System analyses the new message.

5	-	6	System identifies which previous preference has changed.
7	-	8	System updates the changed preference while retaining preferences that were not modified.
9	-	10	System performs UC03 - Obtain Product Recommendations using the revised preferences.
11	Customer reviews the updated recommendations.	12	System retains the revised preferences for subsequent conversation.

Alternative Flows

Table 49: UC06 – Alternative Flows

Step	Branching Action - User	Branching Action - System
A1	Customer supplies a preference that conflicts with an earlier preference.	System prioritises the customer's most recent explicit preference.
A2	Customer rejects a previous recommendation or requests an alternative.	System uses that feedback when generating the next recommendations.
A3	Customer's refinement is unclear.	System performs UC02 - Clarify Shopping Preferences.
A4	Customer changes only one preference.	System retains all other relevant previously supplied preferences.

UC07 - Manage Shopping Cart

Table 50: UC08 – Manage Shopping Cart

Use Case	UC07 - Manage Shopping Cart
Goal	Allow the customer to add recommended products to their shopping cart, review the current cart contents, and remove products they no longer wish to purchase.
Preconditions	The ABC Fashion web application is available to the customer. At least one product has been displayed to the customer before it can be added to the cart.
Success End Condition	The shopping cart reflects the customer's requested changes and the updated cart remains available for the current shopping session.
Failed End Condition	The requested cart action cannot be completed because the selected product is unavailable or a technical error occurs. The customer is informed of the problem

Primary Actors	Customer
Secondary Actors	None
Trigger	The customer selects a shopping-cart action, such as adding a displayed product, viewing the cart, or removing an existing cart item.

Description / Main Success Scenario

Table 51: UC07 – Descriptions and Success Scenarios

Step	User Action	Step	System Response
1	Customer views a displayed recommended product.	2	System displays the product and its available shopping actions.
3	Customer selects Add to Cart.	4	System checks the current availability of the selected product.
5	-	6	System adds the selected product to the shopping cart.
7	Customer opens or reviews the shopping cart.	8	System displays the current shopping cart contents, including the newly added product.
9	Customer continues shopping or accesses the cart.	10	System retains the updated cart for the current shopping session.

Alternative Flows

Table 52: UC07 – Alternative Flows

Step	Branching Action - User	Branching Action - System
A1	The customer selects a product already contained in the cart and chooses to remove it.	The system removes the selected product and updates the shopping cart.
A2	The customer attempts to add a product whose availability has changed.	The system prevents the product from being added and informs the customer that it is unavailable.
A3	The customer attempts to perform a cart action while a technical error occurs.	The system informs the customer that the action could not be completed and keeps the previous cart state where possible.
A4	The customer opens the shopping cart when it contains no products.	The system displays an empty-cart state and allows the customer to continue shopping.